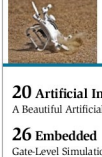
electronics FOR YOU PLUS

FEBRUARY 2016
VOL. 04 NO. 10
ISSN 2454-4426

Next Issue
- Intuitive Gesture Control
- Test and Measurement Equipment
- Buying Noise Meters

Contents



TECH FOCUS
38 | Teach Your Drones To Do More Than Fly

BUYERS' GUIDE
64 | Choosing The Right Bench Power Supply



20 Artificial Intelligence
A Beautiful Artificial Mind

26 Embedded
Gate-Level Simulations: An Increasing Trend

30 Tech Focus
A Sunrise Peppered With Drones

44 Innovation
Diabeto: A 360-Degree Diabetes-Management Solution

48 Telecom
5G: The Next-Generation Network

56 Test & Measurement
The Latest In Scopes

73 Manufacture
Wearable Devices: Essential Inputs for Design Engineers (Part 2 of 2)

76 EFY Plus DVD
Software And Tools To Enrich Your Digital Electronic Utilities

78 Event
India Electronics Week 2016: The New-Age Electronics Show

87 Make in India
Market Survey: Keeping An Eye on India's Surveillance Industry

eStyle

112 Buyers' Guide: Buying A 127cm Flat-Panel TV

114 Do-It-Yourself: Ten Things You Can Do With Your Old Android Device

Regulars

06 Feedback
08 Q&A
10 Useful Websites
14 Tech News
84 Make in India: Industry News
92 New Products
110 First Look
115 Business Pages Ads
126 Electronics Mart Ads
130 Product Categories Index + Attractions During 2016
131 Advertisers' Index

DO-IT-YOURSELF



95 Making Arduino Shields Using Fritzing
97 Lossless Image Compression Using MATLAB
98 RGB Colour Generator
100 Infrared Motion-Sensing Relay Switch
102 PIN Diode Based Fire Sensor
104 Fridge Temperature And Humidity Indicator
106 High-Impedance Audio Buffer With JFET
108 Plus-Minus 5V Supply From 9V Battery

EFY Plus DVD

Signal Processing With Sonic Visualiser I
R: A Data-Analysis And Statistical-Computing Tool ...IV
Let Us Learn how to Program 8085 VII

Interviews

70 CERTIFICATIONS:
"We can even create our own medical device and test it on our neighbour" — Kalyan Varma, vice president - Business Stream Products at TUV Rheinland - TUV Rheinland (India) Pvt Ltd

72 MICROCONTROLLERS:
"Most connected devices are nodes at the last centimetre of networks" — Sanjay Gupta, director, Automotive BU, NXP Semiconductors

EDITOR	: RAMESH CHOPRA
EDITORIAL CORRESPONDENCE	: EDITORIAL SECRETARY PHONE: 011-26810601; E-MAIL: EDITS@EFY.IN (TECHNICAL QUERIES: EFYLAB@EFY.IN)
SUBSCRIPTIONS & MISSING ISSUES	: PHONE: 011-26810601 OR 02 OR 03 E-MAIL: SUPPORT@EFY.IN
BACK ISSUES, BOOKS, CDs, PCBS ETC.	: KITS'N'SPARES, NEW DELHI PHONE: 011-26371661, 26371662 E-MAIL: INFO@KITSNSPARES.COM
EXCLUSIVE NEWSSTAND DISTRIBUTOR	: IBH BOOKS & MAGAZINES DISTRIBUTORS LTD UNIT NO.10, BEZZOLA COMPLEX, OPP. SUMAN NAGAR, SION TROMBAY ROAD, CHEMBUR, MUMBAI 400 071 PHONE: 022-40497401/02
ADVERTISEMENTS	
NEW DELHI (HEAD OFFICE)	: PH: 011-26810601 02 03 E-MAIL: EFYENQ@EFY.IN
MUMBAI	: PH: 022-24950047, 24928520 E-MAIL: EFYMUM@EFY.IN
BENGALURU	: PH: 080-25260394, 25260023 E-MAIL: EFYBLR@EFY.IN
CHENNAI	: PH: 044-42994363 E-MAIL: EFYENQ@EFY.IN
HYDERABAD	: PH: 040-67172633 E-MAIL: EFYENQ@EFY.IN
PUNE	: PH: 08800295610, 09870682995 E-MAIL: EFYPUNE@EFY.IN
GUJARAT	: PH: 079-61344948 E-MAIL: EFYAHD@EFY.IN
CHINA	: POWER PIONEER GROUP INC. PH: (86 755) 83729797, (86) 13923802595 E-MAIL: POWERPIONEER@EFY.IN
JAPAN	: TANDEM INC., PH: 81-3-3541-4166 E-MAIL: TANDEM@EFY.IN
SINGAPORE	: PUBLICITAS SINGAPORE PTE LTD PH: +65-6836 2272 E-MAIL: PUBLICITAS@EFY.IN
TAIWAN	: J.K. MEDIA, PH: 886-2-87726780 EXT. 10 E-MAIL: JKMEDIA@EFY.IN
UNITED STATES	: E & TECH MEDIA PH: +1 860 536 6677 E-MAIL: VERONIQUELAMARQUE@GMAIL.COM

Printed, published and owned by Ramesh Chopra. Printed at International Print-o-Pack Ltd, C-46 to C-11, Hosiery Complex, Phase-II Extension, NOIDA-201 305, Gautam Budh Nagar, Uttar Pradesh, on the first day of each month and published from D-87/1, Okhla Industrial Area, Phase-1, New Delhi 110 020. Copyright 2016. All rights reserved throughout the world. Reproduction of any material from this magazine in any manner without the written permission of the publisher is prohibited. Although every effort is made to ensure accuracy, no responsibility whatsoever is taken for any loss due to publishing errors. Articles that cannot be used are returned to the authors if accompanied by a self-addressed and sufficiently stamped envelope. But no responsibility is taken for any loss or delay in returning the material. EFY will not be responsible for any wrong claims made by an advertiser. Disputes, if any, will be settled in a New Delhi court only.

SUBSCRIPTION RATES

Period	Newsstand Price (₹)	You Pay (₹)	Digital (₹)	Overseas Digital	Overseas Print
Two Year	2400	1800	750	US$ 15	—
One Year	1200	960	280	US$ 5	US$ 120

Please send payments only in favour of **EFY Enterprises Pvt Ltd**